interpreted the large constant term, m_0c^2, to be part of the total energy of the body, an intrinsic energy known as the "rest energy."

Let us follow out the consequences of assuming, with Einstein, that *the energy of a body always equals* mc^2. As an interesting result, we shall find the formula (3.1) for the variation of mass with speed, which we have merely assumed up to now. We start with the body at rest, when its energy is m_0c^2. Then we apply a force to the body, which starts it moving and gives it kinetic energy; therefore, since the energy has increased, the mass has increased—this is implicit in the original assumption. So long as the force continues, the energy and the mass both continue to increase. We have already seen (Chapter 13*) that the rate of change of energy with time equals the force times the velocity, or

$$\frac{dE}{dt} = \mathbf{F} \cdot \mathbf{v}. \tag{3.13}$$

We also have (Chapter 9*, Eq. 9.1) that $F = d(mv)/dt$. When these relations are put together with the definition of E, Eq. (3.13) becomes

$$\frac{d(mc^2)}{dt} = \mathbf{v} \cdot \frac{d(m\mathbf{v})}{dt}. \tag{3.14}$$

We wish to solve this equation for m. To do this we first use the mathematical trick of multiplying both sides by $2m$, which changes the equation to

$$c^2(2m)\frac{dm}{dt} = 2m\mathbf{v} \cdot \frac{d(m\mathbf{v})}{dt}. \tag{3.15}$$

We need to get rid of the derivatives, which can be accomplished by integrating both sides. The quantity $(2m)dm/dt$ can be recognized

* of *The Feynman Lectures on Physics*, vol. I.